



Caringbah High School

Year 12 Mathematics Extension 1 Assessment Task 2 2022

General Instructions

- Reading time – 5 minutes
- Working time – 75 minutes
- Write using black or blue pen
- NESA-approved calculators may be used
- A reference sheet is provided separately
- In Questions 6-10, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 60

Section I 5 marks

Attempt Questions 1-5

Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 55 marks

Attempt Questions 6-10

Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Section I	Section II					Total	
Q1-5	Q6	Q7	Q8	Q9	Q10		
/5	/11	/11	/11	/11	/11	/60	%

Section I -5 marks

Attempt Questions 1-5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5

1 If $\underline{u} = 2\underline{i} + 3\underline{j}$, the unit vector in the direction of $\underline{u}$ is:

A) $\frac{1}{13}(2\underline{i} + 3\underline{j})$

B) $\frac{1}{5}(2\underline{i} + 3\underline{j})$

C) $\frac{1}{\sqrt{13}}(2\underline{i} + 3\underline{j})$

D) $\sqrt{13}(2\underline{i} + 3\underline{j})$

2 What is the derivate of $\tan^{-1}\left(\frac{x}{2}\right)$?

A) $\frac{1}{2(4+x^2)}$

B) $\frac{1}{4+x^2}$

C) $\frac{2}{4+x^2}$

D) $\frac{4}{4+x^2}$

3 Which expression is equal to $\int \cos^2 4x \, dx$?

A) $\frac{-\sin^3 4x}{12} + C$

B) $\frac{\sin^3 4x}{12} + C$

C) $\frac{1}{2}\left(x + \frac{1}{8}\sin 8x\right) + C$

D) $\frac{1}{2}\left(x - \frac{1}{8}\sin 8x\right) + C$

Section I continues on page 3

4 Which of the following is a primitive of $2x(3x^2 - 1)^4$?

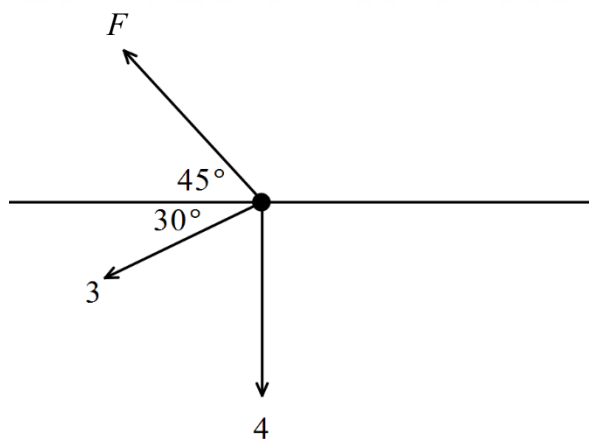
A) $\frac{1}{15}(3x^2 - 1)^5$

B) $\frac{3}{5}(3x^2 - 1)^5$

C) $\frac{2x}{5}(3x^2 - 1)^5$

D) $\frac{2x}{15}(3x^2 - 1)^5$

5 The diagram below shows a mass being acted upon by a number of forces whose magnitudes are labelled. All forces are in newtons, and the system is in equilibrium.



The value of F is:

A) $\frac{\sqrt{2}}{2}(8 + 3\sqrt{3})$

B) $\frac{11\sqrt{2}}{2}$

C) $\frac{3\sqrt{2}}{2}$

D) 7

End of Section I

Section II

55 marks

Attempt Questions 6–10

Allow about 67 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (11 marks)

Marks

- a) Given the vectors $\underline{a} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $\underline{b} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, find:
- | | |
|--|---|
| i) $\underline{b} - 2\underline{a}$ | 1 |
| ii) $ 3\underline{b} $ | 1 |
| iii) $\underline{a} \cdot \underline{b}$ | 1 |
| iv) the angle between the two vectors, correct to the nearest minute | 2 |
| v) a vector that is perpendicular to $\underline{a}$ | 1 |
- b) Differentiate, expressing in simplest form:
- | | |
|------------------------|---|
| i) $\sin^{-1}(4x)$ | 1 |
| ii) $\cos^{-1}(\ln x)$ | 2 |
| iii) $e^x \tan^{-1} x$ | 2 |

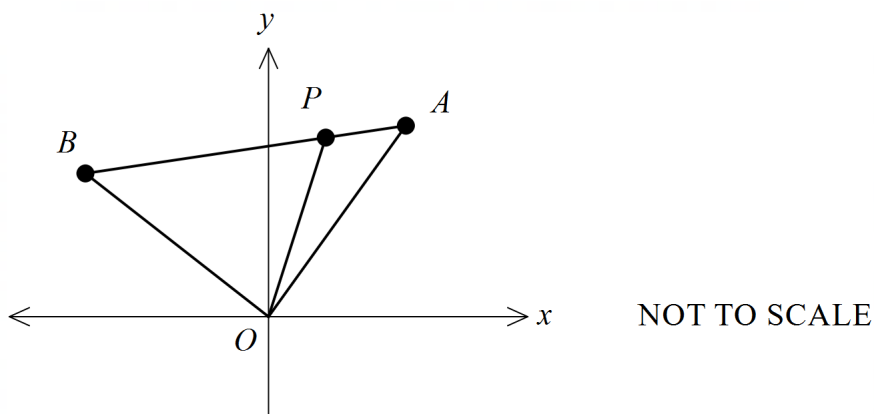
End of Question 6

Question 7 (11 marks)

Marks

- a) Consider two position vectors $\overrightarrow{OA} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\overrightarrow{OB} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$.

Let P be a point on the line passing through A and B such that $\overrightarrow{AP} = m\overrightarrow{AB}$.



- i) Write an expression for $\overrightarrow{AB}$ 1
- ii) Show that the position vector $\overrightarrow{OP} = \begin{bmatrix} 3-7m \\ 4-m \end{bmatrix}$ 2
- iii) Find the value of m if P lies on the line $y = 2x$ 2

- b) Find:

- i) $\int \frac{dx}{2+x^2}$ 1
- ii) $\int \frac{1}{\sqrt{9-25x^2}} dx$ 2

- c) Find the area bounded by the curve $y = \sin^2 x$ and the x -axis for $0 \leq x \leq \pi$ 3

End of Question 7

Question 8 (11 marks)**Marks**

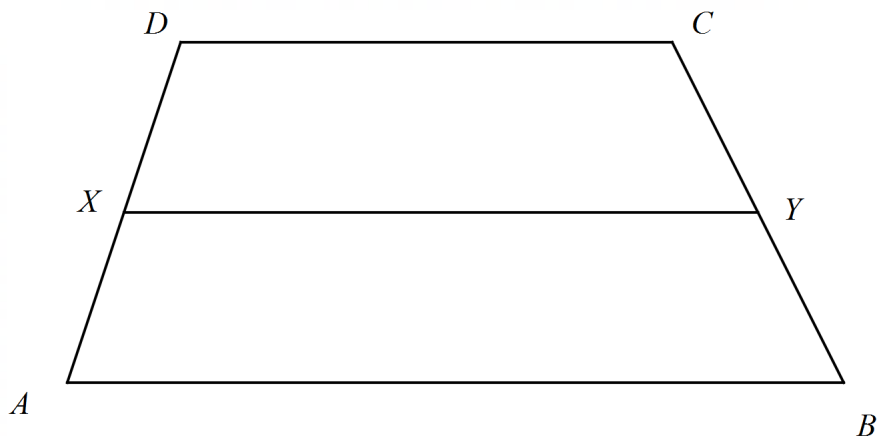
- a) Evaluate $\int \tan^2 x \, dx$ **1**
- b) Given $f(x) = \ln(\sec x + \tan x)$
- (i) Show that $f'(x) = \sec x$ **2**
- (ii) Hence, show that $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec x \, dx = \ln\left(\frac{2+\sqrt{3}}{\sqrt{3}}\right)$ **2**
- c) Evaluate $\int_{-1}^0 x\sqrt{1+x} \, dx$ using the substitution $u = 1+x$ **3**
- d) Find the projection of $\overrightarrow{AB}$ onto $\underline{v} = 2\underline{i} - 3\underline{j}$ given $A(-1, 3)$ and $B(-2, -6)$. **3**

End of Question 8

Question 9 (11 marks)

Marks

- a)** $ABCD$ is a trapezium with $AB \parallel DC$. X and Y are midpoints of AD and BC respectively.



NOT TO SCALE

- | | | |
|------------|---|----------|
| i) | By letting $\underline{a} = \overrightarrow{AB}$, $\underline{b} = \overrightarrow{DC}$, $\underline{c} = \overrightarrow{CB}$, and $\underline{d} = \overrightarrow{AD}$, show that $\overrightarrow{XY} = \frac{1}{2}(\underline{a} + \underline{b})$ | 3 |
| ii) | Hence, show that $XY \parallel AB$ | 2 |
-
- | | | |
|-------------|---|----------|
| b) | Consider the function $f(x) = \sin^{-1}(x^2)$ | |
| i) | State the domain of $f(x)$ | 1 |
| ii) | Find $f'(x)$ | 1 |
| iii) | Hence, find the minimum value of $y = f(x)$ | 2 |
| iv) | Sketch the graph of $y = f(x)$, labelling all key features | 2 |

End of Question 9

Question 10 (11 marks)**Marks**

a) Find $\int \sin 7x \cos 4x \, dx$ **2**

b) i) Show that $6x - x^2 - 5 = 4 - (x - 3)^2$ **1**

ii) Hence, find $\int \frac{-dx}{\sqrt{6x - x^2 - 5}}$ **2**

c) Using the substitution $x = \cos u$, find $\int_{\frac{1}{2}}^1 \frac{1}{x^2 \sqrt{1 - x^2}} \, dx$ **3**

d) The sum of two unit vectors $\underline{a}$ and $\underline{b}$ is a unit vector (i.e. $\underline{a} + \underline{b}$ is also a unit vector). **3**
Prove using vector properties that

$$|\underline{a} - \underline{b}| = \sqrt{3}$$

End of paper

Mathematics Extension 1

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct ↗

-
- Start ➔ 1. A ☐ B ☐ C ☐ D ☐
Here
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐

Yr 12 Maths Ext 1 AT2 Solutions

Section 1

1. C
2. C
3. C
4. A
5. B

1. $u = 2i + 3j$

$$|u| = \sqrt{2^2 + 3^2}$$
$$= \sqrt{13}$$

$$\hat{u} = \frac{u}{|u|}$$

$$= \frac{1}{\sqrt{13}}(2i + 3j)$$

2. $\frac{d}{dx} \tan^{-1}\left(\frac{x}{2}\right) = \frac{1}{1 + (x/2)^2} \cdot \left(\frac{x}{2}\right)'$

$$= \frac{1}{1 + x^2/4} \cdot \frac{1}{2}$$
$$= \frac{4}{4 + x^2} \cdot \frac{1}{2}$$
$$= \frac{2}{4 + x^2}$$

3. $\int \cos^4 x \, dx = \int \frac{1}{2}(1 + \cos 8x) \, dx$

$$= \frac{1}{2} \left(x + \frac{\sin 8x}{8} \right) + C$$

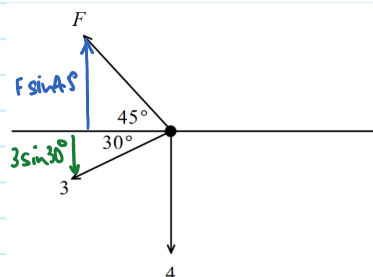
4. $\int 2x(3x^2 - 1)^4 \, dx = \frac{1}{3} \int 6x(3x^2 - 1)^4 \, dx$

$$= \frac{1}{3} \cdot \frac{(3x^2 - 1)^5}{5} + C$$
$$= \frac{(3x^2 - 1)^5}{15} + C$$

5. $F \sin 45^\circ = 3 \sin 30^\circ + 4$

$$\frac{F}{\sqrt{2}} = \frac{3}{2} + 4$$

$$F = \frac{11\sqrt{2}}{2}$$



Section 2

Question 6

$$(a) \quad \underline{a} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad \underline{b} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$(i) \quad \underline{b} - 2\underline{a} = \begin{bmatrix} -2 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 4 \\ 1 \end{bmatrix} \\ = \begin{bmatrix} -10 \\ -1 \end{bmatrix}$$

✓ correct answer

$$(ii) \quad |3\underline{b}| = 3|\underline{b}| \\ = 3\sqrt{(-2)^2 + 3^2} \\ = 3\sqrt{13} \quad \text{or } \sqrt{117}$$

✓ correct answer (should be simplified but do not penalise)

$$(iii) \quad \underline{a} \cdot \underline{b} = 4 \times -2 + 1 \times 3 \\ = -5$$

✓ correct answer

$$(iv) \quad \underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta \\ -5 = \sqrt{17} \times \sqrt{13} \times \cos \theta \\ \cos \theta = \frac{-5}{\sqrt{221}}$$

✓ Substitution into dot product formula

$$\theta = 109^\circ 39' \text{ (arst min)}$$

✓ correct answer

$$(v) \quad \text{We want } \underline{u} = \begin{bmatrix} x \\ y \end{bmatrix} \text{ such that } \underline{u} \perp \underline{a}$$

$$\text{i.e. } \underline{u} \cdot \underline{a} = 0$$

$$4x + y = 0$$

$$y = -4x$$

$$\therefore \text{any vector } \underline{u} = \lambda \begin{bmatrix} 1 \\ -4 \end{bmatrix} \text{ where } \lambda \text{ is a scalar}$$

✓ Correct answer, presumably $\begin{bmatrix} 1 \\ -4 \end{bmatrix}$

$$(b) (i) \quad \frac{d}{dx} [\sin^{-1}(4x)] = \frac{1}{\sqrt{1-(4x)^2}} \cdot (4x)' \\ = \frac{4}{\sqrt{1-16x^2}}$$

✓ correct answer

$$(ii) \quad \frac{d}{dx} [\cos^{-1}(\ln x)] = \frac{-1}{\sqrt{1-(\ln x)^2}} \cdot (\ln x)' \\ = \frac{-1}{x\sqrt{1-(\ln x)^2}}$$

✓ correct progress

✓ correct answer as single fraction

$$(iii) \quad \frac{d}{dx} [e^x \tan^{-1} x] = (e^x)' \tan^{-1} x + e^x (\tan^{-1} x)' \\ = e^x \tan^{-1} x + e^x \frac{1}{1+x^2} \\ = e^x \tan^{-1} x + \frac{e^x}{1+x^2} \\ = e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right)$$

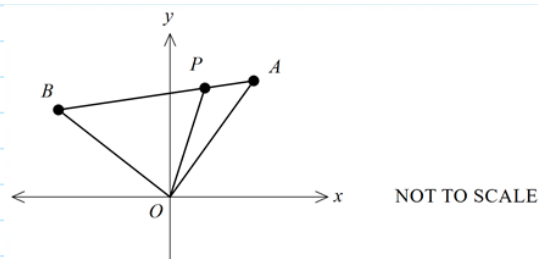
✓ correct use of product rule

✓ correct answer (no need to factorise)

Question 7

$$(a) \quad \vec{OA} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad \vec{OB} = \begin{bmatrix} -4 \\ 3 \end{bmatrix} \quad \vec{AP} = m \vec{AB}$$

$$(i) \quad \vec{AB} = \begin{bmatrix} -4 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -7 \\ -1 \end{bmatrix} \quad \checkmark \text{ correct answer}$$



$$(ii) \quad \vec{OP} = \vec{OA} + \vec{AP} = \vec{OA} + m \vec{AB} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} + m \begin{bmatrix} -7 \\ -1 \end{bmatrix} = \begin{bmatrix} 3-7m \\ 4-m \end{bmatrix} \quad \checkmark \text{ correct addition of vectors}$$

$$(iii) \quad \text{Sub } \vec{OP} \text{ in } y=2x \\ 4-m = 2(3-7m) \\ = 6-14m \\ 13m = 2 \\ \therefore m = \frac{2}{13} \quad \checkmark \text{ correct 'show'}$$

$$(b) (i) \quad \int \frac{dx}{2+x^2} = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + C \quad \checkmark \text{ correct substitution \& linear equation}$$

$$(ii) \quad \int \frac{1}{\sqrt{9-25x^2}} dx = \frac{1}{5} \int \frac{5}{\sqrt{3^2-(5x)^2}} dx \quad \checkmark \text{ correct answer}$$

$$= \frac{1}{5} \sin^{-1}\left(\frac{5x}{3}\right) + C \quad \checkmark \text{ correct progress}$$

$$(c) \quad \text{Area} = \int_0^{\pi} \sin^2 x \, dx = \int_0^{\pi} \frac{1}{2} (1 - \cos 2x) \, dx \quad \checkmark \text{ correct expression for } \sin^2 x$$

$$= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi} \quad \checkmark \text{ correct integration}$$

$$= \frac{1}{2} \left(\pi - \frac{1}{2} \sin 2\pi - \left(0 - \frac{1}{2} \sin 0 \right) \right)$$

$$= \frac{\pi}{2} \quad \checkmark \text{ correct answer}$$

Question 8

$$(a) \int \tan^2 x \, dx = \int \sec^2 x - 1 \, dx$$

$$= \tan x - x + C$$

✓ correct answer

$$(b) (i) f(x) = \ln(\sec x + \tan x)$$

$$f'(x) = \frac{1}{\sec x + \tan x} \cdot (\sec x + \tan x)'$$

$$\downarrow$$

$$\frac{d}{dx} \sec x = \frac{d}{dx} (\cos x)^{-1}$$

$$= -1(\cos x)^{-2} \cdot -\sin x$$

$$= \frac{\sin x}{\cos^2 x}$$

$$= \sec x \tan x$$

$$= \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x}$$

$$= \frac{\sec x (\tan x + \sec x)}{\sec x + \tan x}$$

$$= \sec x$$

✓ correct application of chain rule

✓ correct 'show' (simplify)

$$(ii) \text{ From (i), } \frac{d}{dx} (\ln(\sec x + \tan x)) = \sec x$$

$$\text{then } \int \sec x \, dx = \ln(\sec x + \tan x) + C$$

$$\text{Now, } \int_{\pi/6}^{\pi/3} \sec x \, dx = \left[\ln(\sec x + \tan x) \right]_{\pi/6}^{\pi/3}$$

$$= \ln\left(\sec \frac{\pi}{3} + \tan \frac{\pi}{3}\right) - \ln\left(\sec \frac{\pi}{6} + \tan \frac{\pi}{6}\right)$$

$$= \ln(2 + \sqrt{3}) - \ln\left(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}\right)$$

$$= \ln(2 + \sqrt{3}) - \ln\left(\frac{3}{\sqrt{3}}\right)$$

$$= \ln(2 + \sqrt{3}) - \ln \sqrt{3}$$

$$= \ln\left(\frac{2 + \sqrt{3}}{\sqrt{3}}\right)$$

✓ correct substitution

✓ correct 'show' (simplify)

$$(c) \int_{-1}^0 x \sqrt{1+x} \, dx$$

$$\text{Let } u = 1+x \rightarrow x = u-1$$

$$du = dx$$

$$\begin{cases} x=0, u=1 \\ x=-1, u=0 \end{cases}$$

$$= \int_0^1 (u-1) \cdot u^{1/2} \, du$$

$$= \int_0^1 u^{3/2} - u^{1/2} \, du$$

$$= \left[\frac{u^{5/2}}{5/2} - \frac{u^{3/2}}{3/2} \right]_0^1$$

$$= \frac{2}{5}(1) - \frac{2}{3}(1) - 0$$

$$= -\frac{4}{15}$$

✓ correct substitution

✓ correct integration

✓ correct answer

(d) $\underline{v} = 2\hat{i} - 3\hat{j}$ $A(-1,3)$ & $B(-2,-6)$

$$\vec{AB} = -\hat{i} - 9\hat{j}$$

✓ Correct $\vec{AB}$

$$\text{Proj}_{\underline{u}} \underline{u} = \frac{\underline{u} \cdot \underline{u}}{\underline{u} \cdot \underline{u}} \times \underline{u}$$

$$\text{Proj}_{\underline{v}} \vec{AB} = \frac{(-\hat{i} - 9\hat{j}) \cdot (2\hat{i} - 3\hat{j})}{(2\hat{i} - 3\hat{j}) \cdot (2\hat{i} - 3\hat{j})} \times (2\hat{i} - 3\hat{j})$$

$$= \frac{-1 \times 2 + -9 \times -3}{2^2 + (-3)^2} \times (2\hat{i} - 3\hat{j})$$

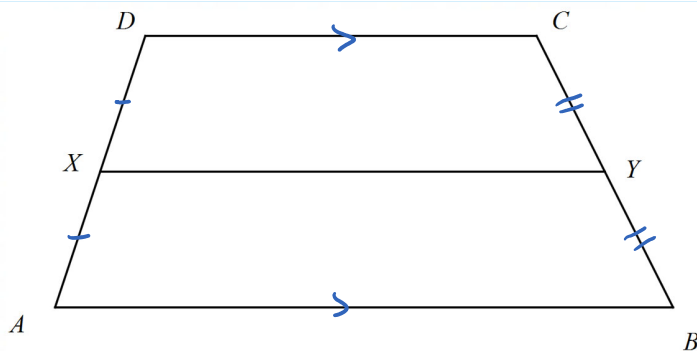
✓ Correct use of projection formula

$$= \frac{25}{13} (2\hat{i} - 3\hat{j})$$

✓ correct answer

Question 9

(a) (i) $\underline{a} = \vec{AB}$, $\underline{b} = \vec{DC}$, $\underline{c} = \vec{CB}$, $\underline{d} = \vec{AD}$



$$\vec{AB} = \vec{AD} + \vec{DC} + \vec{CB}$$

$$\underline{a} = \underline{d} + \underline{b} + \underline{c}$$

$$\underline{d} + \underline{c} = \underline{a} - \underline{b}$$

✓ correct relationship between $\underline{a}, \underline{b}, \underline{c}, \underline{d}$

$$\vec{AB} = \vec{AX} + \vec{XY} + \vec{YB}$$

$$= \frac{1}{2}\vec{AD} + \vec{XY} + \frac{1}{2}\vec{CB}$$

$$\underline{a} = \frac{1}{2}\underline{d} + \vec{XY} + \frac{1}{2}\underline{c}$$

$$= \frac{1}{2}(\underline{c} + \underline{d}) + \vec{XY}$$

$$= \frac{1}{2}(\underline{a} - \underline{b}) + \vec{XY}$$

$$\vec{XY} = \underline{a} - \frac{1}{2}(\underline{a} - \underline{b})$$

$$= \underline{a} - \frac{1}{2}\underline{a} + \frac{1}{2}\underline{b}$$

$$= \frac{1}{2}(\underline{a} + \underline{b})$$

✓ second expression involving $\vec{XY}$

✓ correct 'show'

(ii) Since $AB \parallel DC$, $\underline{b} = \lambda \underline{a}$, λ scalar

✓ correct relationship between $\underline{a}$ & $\underline{b}$

$$\vec{XY} = \frac{1}{2}(\underline{a} + \underline{b})$$

$$= \frac{1}{2}(\underline{a} + \lambda \underline{a})$$

$$= \frac{\lambda + 1}{2} \times \underline{a}$$

$$= \mu \underline{a}, \mu \text{ scalar}$$

✓ correct 'show'

$$\therefore XY \parallel AB$$

(b) $f(x) = \sin^{-1}(x^2)$

(i) D: $-1 \leq x^2 \leq 1$

$\therefore -1 \leq x \leq 1$

✓ Correct answer

(ii) $f'(x) = \frac{1}{\sqrt{1-(x^2)^2}} \cdot (x^2)'$
 $= \frac{2x}{\sqrt{1-x^4}}$

✓ Correct answer

(iii) Max/min value when $f'(x) = 0$

$f'(x) = \frac{2x}{\sqrt{1-x^4}} = 0$

$2x = 0$

$x = 0$

✓ Correct $x=0$

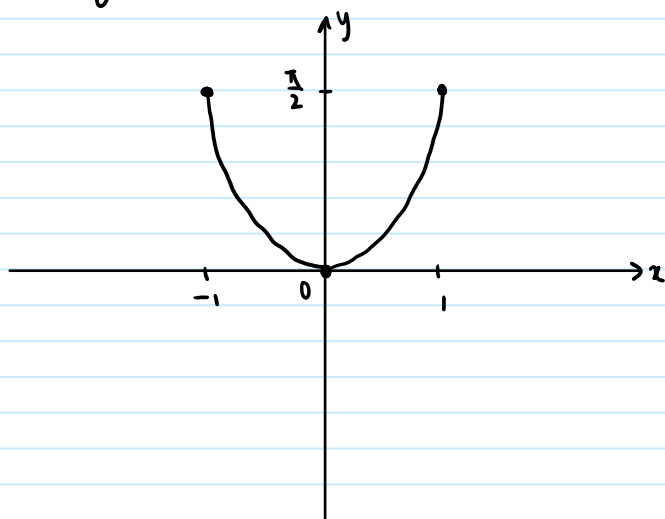
When $x=0$, $y = f(0) = \sin^{-1} 0 = 0$

x	$-1/2$	0	$1/2$
$f'(x)$	-1.03	0	1.03
	↓	→	↑

✓ Sub & test for min

$\therefore$ minimum value is 0.

(iv) $y = \sin^{-1}(x^2)$



✓ Correct min SP

✓ Correct endpoints

Note: Should be vert. tangent at $x = \pm 1$.

Question 10

$$\begin{aligned}
 (a) \quad \int \sin 7x \cos 4x \, dx &= \int \frac{1}{2} (\sin(7x+4x) + \sin(7x-4x)) \, dx \\
 &= \frac{1}{2} \int \sin 11x + \sin 3x \, dx \\
 &= \frac{1}{2} \left(-\frac{\cos 11x}{11} + -\frac{\cos 3x}{3} \right) + C \\
 &= -\frac{1}{2} \left(\frac{\cos 3x}{3} + \frac{\cos 11x}{11} \right) + C
 \end{aligned}$$

✓ correct 'product to sum' formula

✓ correct integration

$$\begin{aligned}
 (b) \quad (i) \quad \text{RHS} &= 4 - (x-3)^2 \\
 &= 4 - (x^2 - 6x + 9) \\
 &= 4 - x^2 + 6x - 9 \\
 &= 6x - x^2 - 5 \\
 &= \text{LHS}
 \end{aligned}$$

✓ correct 'show'

$$\begin{aligned}
 (ii) \quad \int \frac{-dx}{\sqrt{6x-x^2-5}} &= \int \frac{-1}{\sqrt{4-(x-3)^2}} \, dx \\
 &= -\sin^{-1} \left(\frac{x-3}{2} \right) + C \\
 &= \cos^{-1} \left(\frac{x-3}{2} \right) + C
 \end{aligned}$$

✓ correct progress

✓ correct integral

$$(c) \quad \int_{1/2}^1 \frac{1}{x^2 \sqrt{1-x^2}} \, dx$$

$$\begin{aligned}
 \text{Let } x &= \cos u \\
 dx &= -\sin u \, du \\
 \begin{cases} x=1, & u=0 \\ x=1/2, & u=\pi/3 \end{cases}
 \end{aligned}$$

✓ correct substitution

$$= \int_{\pi/3}^0 \frac{1}{\cos^2 u \sqrt{1-\cos^2 u}} \cdot -\sin u \, du$$

$$= \int_0^{\pi/3} \frac{\cancel{\sin u}}{\cos^2 u \sqrt{\cancel{1-\cos^2 u}}} \, du$$

$$= \int_0^{\pi/3} \sec^2 u \, du$$

✓ correct simplification

$$= [\tan u]_0^{\pi/3}$$

$$= \tan \frac{\pi}{3} - \tan 0$$

$$= \sqrt{3} - 0$$

$$= \sqrt{3}$$

✓ correct answer

(11) RTP: $|\underline{a} - \underline{b}| = \sqrt{3}$ given $|\underline{a}| = |\underline{b}| = |\underline{a} + \underline{b}| = 1$

$$\underline{u} \cdot \underline{u} = |\underline{u}|^2$$

✓ Recognition of need to use
 $\underline{u} \cdot \underline{u} = |\underline{u}|^2$

Consider $|\underline{a} + \underline{b}|^2 = (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b})$

$$1^2 = \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b}$$

$$1 = |\underline{a}|^2 + 2\underline{a} \cdot \underline{b} + |\underline{b}|^2$$

$$= 1^2 + 2\underline{a} \cdot \underline{b} + 1^2$$

$$2\underline{a} \cdot \underline{b} = -1$$

✓ Correct expression for $2\underline{a} \cdot \underline{b}$

Now, $|\underline{a} - \underline{b}|^2 = (\underline{a} - \underline{b}) \cdot (\underline{a} - \underline{b})$

$$= \underline{a} \cdot \underline{a} - \underline{a} \cdot \underline{b} - \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b}$$

$$= |\underline{a}|^2 - 2\underline{a} \cdot \underline{b} + |\underline{b}|^2$$

$$= 1^2 - 2\underline{a} \cdot \underline{b} + 1^2$$

$$= 2 - (-1)$$

$$= 3$$

$$\therefore |\underline{a} - \underline{b}| = \sqrt{3}$$

✓ Correct 'show'